

# FORTESIA FILM COATED TABLET 100MG

Sitagliptin (as Phosphate Monohydrate) 100mg

### DESCRIPTION

Beige, round, biconvex, film coated tablet with “100” on one side and scored on the other. Tablet can be split into half when necessary.

### THERAPEUTIC CLASS

Fortesia is an orally-active, potent, and highly selective inhibitor of the dipeptidyl peptidase 4 (DPP-4) enzyme for the treatment of type 2 diabetes. The DPP-4 inhibitors are a class of agents that act as incretin enhancers. By inhibiting the DPP-4 enzyme, sitagliptin increases the levels of two known active incretin hormones, glucagon-like peptide-1 (GLP-1) and glucose-dependent insulinotropic polypeptide (GIP). The incretins are part of an endogenous system involved in the physiologic regulation of glucose homeostasis. When blood glucose concentrations are normal or elevated, GLP-1 and GIP increase insulin synthesis and release from pancreatic beta cells. GLP-1 also lowers glucagon secretion from pancreatic alpha cells, leading to reduced hepatic glucose production. This mechanism is unlike the mechanism seen with sulfonylureas; sulfonylureas cause insulin release even when glucose levels are low, which can lead to sulfonylurea-induced hypoglycaemia in patients with type 2 diabetes and in normal subjects. Sitagliptin is a potent and highly selective inhibitor of the enzyme DPP-4 and does not inhibit the closely-related enzymes DPP-8 or DPP-9 at therapeutic concentrations. Sitagliptin differs in chemical structure and pharmacological action from GLP-1 analogues, insulin, sulfonylureas or meglitinides, biguanides, peroxisome proliferator-activated receptor gamma (PPAR $\gamma$ ) agonists, alpha-glucosidase inhibitors, and amylin analogues.

### COMPOSITION

#### Active Ingredients

Each film coated of Fortesia Tablet contains 128.50 mg sitagliptin phosphate monohydrate contains 100 mg of sitagliptin.

### MECHANISM OF ACTION

Fortesia is a member of a class of oral anti-hyperglycaemic agents called dipeptidyl peptidase 4 (DPP-4) inhibitors, which improve glycaemic control in patients with type 2 diabetes by enhancing the levels of active incretin hormones. Incretin hormones, including glucagon like peptide-1 (GLP1) and glucose dependent insulinotropic polypeptide (GIP) are released by the intestine throughout the day and levels are increased in response to a meal, The incretins are part of an endogenous system involved in the physiologic regulation of glucose homeostasis. When blood glucose concentrations are normal or elevated, GLP-1 and GIP increase insulin synthesis and release from pancreatic beta cells by intracellular signalling pathways involving cyclic AMP. Treatment with GLP-1 or with DPP-4 inhibitors in animal models of type 2 diabetes has been demonstrated to improve beta cell responsiveness to glucose and stimulate insulin biosynthesis and release. With higher insulin levels, tissue glucose uptake is enhanced. In addition, GLP-1 lowers glucagon secretion from pancreatic alpha cells. Decreased glucagon concentrations, along with higher insulin levels, lead to reduced hepatic glucose production, resulting in a decrease in blood glucose levels. The effects of GLP-1 and GIP are glucose dependent such that when blood glucose concentrations are low, stimulation of insulin

release and suppression of glucagon secretion by GLP-1 are not observed. For both GLP-1 and GIP, stimulation of insulin release is enhanced as glucose rises above normal concentrations. Further, GLP-1 does not impair the normal glucagon response to hypoglycaemia. The activity of GLP -1 and GIP is limited by the DPP-4 enzyme, which rapidly hydrolyses the incretin hormones to produce inactive products. Sitagliptin prevents the hydrolysis of incretin hormones by DPP-4 thereby increasing plasma concentrations of the active forms of GLP-1 and GIP. By enhancing active incretins levels, sitagliptin increases insulin release and decreases glucagon levels in a glucose dependent manner. In patients with type 2 diabetes with hyperglycaemia, these changes in insulin and glucagon levels lead to lower haemoglobin A<sub>1c</sub> (HBA<sub>1c</sub>) and lower fasting and postprandial glucose concentrations. The glucose dependent mechanism of sitagliptin is distinct from the mechanism of sulfonylureas, which increase secretion even when glucose levels are low and can lead to hypoglycaemia in patients with type 2 diabetes and in normal subjects. Sitagliptin is a potent and highly selective inhibitor of the enzyme DPP-4 and does not inhibit the closely related enzymes DPP-8 or DPP-9 at therapeutic concentrations.

### PHARMACODYNAMICS

#### General

In patients with type 2 diabetes, administration of single oral doses of Fortesia Tablet leads to inhibition of DPP-4 enzyme activity for a 24-hour period, resulting in a 2-to-3-fold increase in circulating levels of active GLP-1 and GIP, increased plasma levels of insulin and C-peptide, decreased glucagon concentrations, reduced fasting glucose and reduced glucose excursion following an oral glucose load or a meal.

Sitagliptin alone increased only active GLP-1 concentrations, reflecting inhibition of DPP-4, whereas metformin alone increased active and total GLP-1 concentrations to a similar extent.

### PHARMACOKINETICS

#### Absorption

The absolute bioavailability of sitagliptin is approximately 87%. Since co-administration of a high fat meal with Fortesia Tablet had no effect on the pharmacokinetics, Fortesia may be administered with or without food.

#### Distribution

The mean volume of distribution of steady state following a single 100mg intravenous dose of sitagliptin to healthy subjects is approximately 198 litres. The fraction of sitagliptin reversibly bound to plasma is low 38%.

#### Metabolism

Sitagliptin is primarily eliminated unchanged in urine, and metabolism is a minor pathway. Approximately 79% of sitagliptin is excreted unchanged in the urine.

Following a [<sup>14</sup>C] sitagliptin oral dose, approximately 16% of the radioactivity was excreted as metabolites of sitagliptin. Six metabolites were detected at trace levels and are not expected to contribute to the plasma DPP-4 inhibitory activity of sitagliptin. In vitro studies indicated that the primary enzyme responsible for the limited metabolism of sitagliptin was CYP3A4, with contribution from CYP2C8.

### Elimination

Following administration of an oral [<sup>14</sup>C] sitagliptin dose, approximately 100% of the administered radioactivity was eliminated in faeces (13%) or urine (87%) within one week of dosing. The apparent terminal t<sub>½</sub> following a 100mg oral dose of sitagliptin was approximately 12.4 hours and renal clearance was approximately 350mL/min.

Elimination of sitagliptin occurs primarily via renal excretion and involves active tubular secretion. Sitagliptin is a substrate for human organic anion transporter-3 (hOAT -3) which may be involved in the renal elimination of sitagliptin. The clinical relevance of hOAT-3 in sitagliptin transport has not been established. Sitagliptin is also a substrate of p-glycoprotein which may also be involved in mediating the renal elimination of sitagliptin. However, cyclosporine, a p-glycoprotein inhibitor, did not reduce clearance of sitagliptin.

### Renal impairment

Sitagliptin was modestly removed by haemodialysis (13.5% over a 3- to 4-hour haemodialysis session starting 4 hours post dose). To achieve plasma concentrations of sitagliptin similar to those in patients with normal renal function, lower dosages are recommended in patients with moderate and severe renal insufficiency, as well as in ESRD patients requiring haemodialysis.

### Hepatic impairment

No dosage adjustment for Fortesia Tablet is necessary for patients with mild or moderate hepatic insufficiency. There is no clinically experience in patients with severe hepatic insufficiency (Child- Pugh score >9). However, because sitagliptin is primarily renally eliminated, severe hepatic insufficiency is not expected to affect the pharmacokinetic of sitagliptin.

### Elderly

No dosage adjustment is required based on age. Elderly subjects (65 to 85 years) had approximately 19% higher plasma concentrations of sitagliptin compared to younger subjects.

### Paediatric

No studies with sitagliptin have been performed in paediatric patients <10 years of age.

### Gender

No dosage adjustment is necessary based on gender.

### Race

No dosage adjustment is necessary based on race.

### Body Mass Index (BMI)

No dosage adjustment is necessary based on BMI.

### Type 2 diabetes

The pharmacokinetic of sitagliptin in patients with type 2 diabetes are generally similar to those in healthy subjects.

### INDICATIONS

#### Monotherapy

Fortesia Tablet is indicated as an adjunct to diet and exercise to improve glycaemic control in patients with type 2 diabetes mellitus.

Combination with metformin

Fortesia is indicated in patients with type 2 diabetes mellitus to improve glycaemic control in combination with metformin as initial therapy or when the single agent alone, with diet and exercise does not provide adequate glycaemic control. Initial combination therapy or maintenance of combination therapy may not be appropriate for all patients. These management options are left to the discretion of the health care provider.

Combination with a sulphonylurea

Fortesia is indicated in patients with type 2 diabetes mellitus to improve glycaemic control in combination with sulphonylurea when treatment with maximal tolerated dose of sulphonylurea alone, with diet and exercise, does not provide adequate glycaemic control and when metformin is inappropriate due to contraindications or intolerance.

Combination with metformin and sulphonylurea

Fortesia is indicated in patients with type 2 diabetes mellitus to improve glycaemic control in combination with metformin and a sulphonylurea when dual therapy with these two agents and with diet and exercise does not provide adequate glycaemic control.

Combination with peroxisome proliferator-activated receptor gamma (PPARγ) agonist

Fortesia is indicated in patients with type 2 diabetes mellitus to improve glycaemic control in combination with a PPARγ agonist (i.e., thiazolidinediones) when diet and exercise, plus the single agent do not provide adequate glycaemic control.

Combination with metformin and a PPARγ agonist

Fortesia is indicated in patients with type 2 diabetes mellitus to improve glycaemic control in combination with metformin and PPARγ agonist (i.e thiazolidinediones) when dual therapy with these agents, with diet and exercise does not provide adequate control.

Combination with insulin

Fortesia is indicated in patients with type 2 diabetes mellitus as an adjunct to diet and exercise to improve glycaemic control in combination with insulin (with or without metformin).

DOSAGE AND ADMINISTRATION

The recommended dose for Fortesia Tablet is 100 mg once daily as monotherapy or as combination therapy with metformin, a sulfonylurea, insulin (with or without metformin), PPARγ agonist (i.e thiazolidinediones) metformin plus a sulfonylurea or metformin plus a PPARγ agonist Fortesia Tablet can be taken with or without food.

When Fortesia Tablet is used in combination with a sulfonylurea or with insulin, a lower dose of sulfonylurea or insulin may be considered to reduce the risk of sulfonylurea or insulin induced hypoglycemia (See PRECAUTIONS, hypoglycemia in combination with a sulfonylurea or with insulin)

Patients with renal impairment  
Because there is a dosage adjustment based upon renal function, assessment of renal function is recommended prior to initiation of Fortesia and periodically thereafter.

For patients with mild renal impairment (estimated glomerular filtration rate [eGFR] ≥60 mL/min/1.73 m2 to < 90 mL/min/1.73 m²), no dosage adjustment for Fortesia is required.

For patients with moderate renal impairment (eGFR ≥ 45 mL/min.1.73 m² to < 60 mL/min.1.73 m²), no dosage adjustment for Fortesia is required.

For patients with moderate renal impairment (eGFR ≥ 30 mL/min/1.73 m² to 45 mL/min/1.73 m²), the dose of Fortesia is 50 mg once daily.

For patients with severe renal impairment (eGFR ≥ 15 mL/min/1.73 m² to < 30 mL/min/1.73 m²) or with end-stage renal disease (ESRD) (eGFR < 15 mL/min/1.73m²), including those requiring hemodialysis or peritoneal dialysis, the dose of Fortesia is 25 mg once daily. Fortesia may be administered without regard to the timing of dialysis.

Pediatric population

Fortesia should not be used in children and adolescents 10 to 17 years of age because of insufficient efficacy. Fortesia has not been studied in pediatric patients under 10 years of age

CONTRAINDICATIONS

Fortesia is contraindicated in patients who are hypersensitive to any components of this product. (See PRECAUTIONS, Hypersensitivity Reactions and SIDE EFFECTS)

PRECAUTIONS

**General:** Fortesia should not be used in patients with type 1 diabetes or for the treatment of diabetic ketoacidosis.

**Pancreatitis:** There have been reports of acute pancreatitis, including fatal and non-fatal haemorrhagic or necrotizing pancreatitis (see SIDE EFFECTS) in patients taking sitagliptin. Patients should be informed of the characteristic symptom of acute pancreatitis; persistent, severe abdominal pain. Resolution of pancreatitis has been observed after discontinuation of sitagliptin. If pancreatitis is suspected Fortesia and other potentially suspect medicinal products should be discontinued.

**Use in Patients with Renal Insufficiency:** Fortesia is renally excreted. To achieve plasma concentrations of Fortesia Tablet similar to those in patients with normal renal function; lower dosages are recommended in patients with moderate and severe renal insufficiency, as well as in ESRD patients requiring haemodialysis or peritoneal dialysis. (See DOSAGE AND ADMINISTRATION, Patients with renal insufficiency)

**Hypoglycaemia in Combination with Sulfonylurea or with Insulin:** As is typical with other antihyperglycaemic agents, hypoglycaemia is expected to be observed when Fortesia was used in combination with insulin or a sulfonylurea (see SIDE EFFECTS). Therefore, to reduce the risk of sulfonylurea- or insulin-induced hypoglycaemia, a lower dose of sulfonylurea or insulin may be considered (see DOSAGE AND ADMINISTRATION)

**Hypersensitivity Reactions:** The reactions include anaphylaxis, angioedema and exfoliative skin conditions including Stevens-Johnson syndrome. If a hypersensitivity reaction is suspected, discontinue Fortesia, assess for other potential causes for the event, and institute alternative treatment for diabetes (See CONTRAINDICATIONS and SIDE EFFECTS)

**Severe and disabling arthralgia:** Consider DPP-4 inhibitors as a possible cause of severe joint pain and discontinue drug if appropriate.

**Bullous Pemphigoid:** Tell patients to report development of blisters or erosions while receiving Fortesia. If bullous pemphigoid is suspected, Fortesia should be discontinued and referral to a dermatologist should be considered for diagnosis and appropriate treatment.

PREGNANCY

There are no adequate and well controlled studies in pregnant women, therefore the safety of Fortesia in pregnant women is not known. Fortesia, like other oral antihyperglycaemic agents is not recommended for use in pregnancy.

NURSING MOTHER

Sitagliptin is secreted in the milk of lactating rats. It is not known whether sitagliptin is secreted in human milk. Therefore, Fortesia should not be used by a woman who is nursing.

PEDIATRIC USE

Fortesia has not been studied in paediatric patients under 10 years of age.

USE IN THE ELDERLY

No dosage adjustment is required based on age. Elderly patients are more likely to have renal impairment; as with other patients, dosage adjustment may be required in the presence of significant renal impairment (see DOSAGE AND ADMINISTRATION, patient with renal impairment)

DRUG INTERACTION

Patients receiving digoxin should be monitored appropriately. No dosage adjustment of digoxin or Fortesia Tablet is recommended.

No dosage adjustment for Fortesia is recommended when co-administered with cyclosporine or other p-glycoprotein inhibitors (e.g ketoconazole).

SIDE EFFECTS

Fortesia was generally well tolerated, however below are some side effects which may encountered

| Side Effects Reaction                                         | Frequency of Side Effects |
|---------------------------------------------------------------|---------------------------|
| <b>Blood and lymphatic system disorders</b>                   |                           |
| thrombocytopenia                                              | Rare                      |
| <b>Immune system disorders</b>                                |                           |
| hypersensitivity reactions including anaphylactic responses   | Frequency not known       |
| <b>Metabolism and nutrition disorders</b>                     |                           |
| hypoglycaemia                                                 | Common                    |
| <b>Nervous system disorders</b>                               |                           |
| headache                                                      | Common                    |
| dizziness                                                     | Uncommon                  |
| <b>Respiratory, thoracic and mediastinal disorders</b>        |                           |
| interstitial lung disease                                     | Frequency not known       |
| <b>Gastrointestinal disorders</b>                             |                           |
| constipation                                                  | Uncommon                  |
| vomiting                                                      | Frequency not known       |
| acute pancreatitis                                            | Frequency not known       |
| fatal and non-fatal haemorrhagic and necrotizing pancreatitis | Frequency not known       |
| <b>Skin and subcutaneous tissue disorders</b>                 |                           |
| pruritus                                                      | Uncommon                  |
| angioedema                                                    | Frequency not known       |

|                                                                |                     |
|----------------------------------------------------------------|---------------------|
| rash                                                           | Frequency not known |
| urticaria                                                      | Frequency not known |
| cutaneous vasculitis                                           | Frequency not known |
| exfoliative skin conditions including Stevens-Johnson syndrome | Frequency not known |
| bullous pemphigoid                                             | Frequency not known |
| <b>Musculoskeletal and connective tissue disorders</b>         |                     |
| arthralgia                                                     | Frequency not known |
| myalgia                                                        | Frequency not known |
| back pain                                                      | Frequency not known |
| arthropathy                                                    | Frequency not known |
| <b>Renal and urinary disorders</b>                             |                     |
| impaired renal function                                        | Frequency not known |
| acute renal failure                                            | Frequency not known |

**EFFECTS ON ABILITY TO DRIVE AND USE MACHINES**  
Fortesia has no or negligible influence on the ability to drive and use machines. However, when driving or using machines, it should be taken into account that dizziness and somnolence have been reported.

In addition, patients should be alerted to the risk of hypoglycaemia when Fortesia is used in combination with a sulphonylurea or with insulin.

**OVERDOSE**  
In the event of an overdose, it is reasonable to employ the usual supportive measures e.g remove unabsorbed material from the gastrointestinal tract, employ clinical monitoring (including obtaining an electrocardiogram and institute supportive therapy if required. Sitagliptin is modestly dialyzable. Prolonged hemodialysis may be considered if clinically appropriate. It is not known if sitagliptin is dialyzable by peritoneal dialysis.

**ROUTE OF ADMINISTRATION**  
Oral.

**SHELF LIFE**  
Please refer to outer package.

**PACK SIZES**  
3 x 10's Alu-PVC/PE/PVDC Blister Pack.

**STORAGE CONDITION**  
Store in a dry place 30°C. Protect from light.  
Keep out of reach of children.  
Jauhkan daripada kanak-kanak.

**PRODUCT REGISTRATION HOLDER**  
**Duopharma Manufacturing (Bangi) Sdn. Bhd.**  
Lot No. 2, 4, 6, 8, 10, Jalan P/7, Section 13,  
Bangi Industrial Estate, 43650,  
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**DATE OF REVISION**  
March 2022

Remarks:  
Leaflet is preliminary drafted in Calibri with Font size of 11 which subject to change to any equivalent font and readable font size.